

Variable Dictionary

Project: UnionSpill — Layer Connectivity Pipeline

Authors: Luis de Azevedo-Gomes & Guilherme Neri

This dictionary covers three datasets produced by the layer connectivity pipeline (Programs/layer_connectivity/). They are used as inputs to 07_layer_spillover.do to estimate within-firm, cross-layer spillover effects of the Súmula 277 reform.

Dataset 1 — totalflows_wide_2007_2011.csv

File: totalflows_wide_2007_2011.csv | **Path:** Data/RAIS_aux/

Unit of observation: establishment | **Rows:** 16,470 firms | **Year pairs covered:** 2007–08, 2008–09, 2009–10, 2010–11

This file contains firm-level total worker flows for each of the four pre-treatment year pairs. A "flow" between years t and $t+1$ counts any worker who was employed at the focal firm in December of year t or December of year $t+1$ (outflows + inflows). The file is constructed in 05_yearly_employers.do and used in 07_layer_spillover.do for the firm-level restricted specification.

1.1 Identifier

Variable	Type	Description
identificad	INTEGER	14-digit establishment identifier (CNPJ-base). Note: stored as integer here; must be read with stringcols(1) in Stata or cast to VARCHAR for joins with other pipeline files that store it as a 14-char string.

1.2 Total Flow Variables

For each year pair AB (07_08, 08_09, 09_10, 10_11), two variables are provided. Flows are symmetric: a worker who left in year A or arrived in year B is counted once. Null values arise when the firm does not appear in one of the two annual RAIS cross-sections (typically newly-entered or exiting firms at the edge of the panel).

Variable	Type	Mean	Max	Nulls	Description
totalflows_AB	DOUBLE	~25.3	~4,149	~722 (07_08)	Raw count of worker flows between the focal firm and all other firms in year pair AB . Equals outflows + inflows, where an outflow is a worker present in December t and absent in December $t+1$ (at this firm), and an inflow is the reverse.
totalflows_pw_AB	DOUBLE	~0.26	2.00	same as above	Per-worker flow rate: $\text{totalflows_AB} / E_{\text{avg}}$, where $E_{\text{avg}} = (\text{December employment in year } t + \text{December employment in year } t+1) / 2$. Values above 1 are possible when flows exceed average employment (high-turnover firms). Used in pre-treatment control bins.

Dataset 2 — firm_layer_outcomes_{layer}.dta

Files: firm_layer_outcomes_edu2.dta | firm_layer_outcomes_gender.dta | firm_layer_outcomes_race.dta

Path: Data/layer_connectivity/ | **Unit:** establishment $\times$ layer $\times$ year | **Years:** 2009–2016

Layer	Layer values	Rows	Unique firms
edu	0_no_hs / 1_hs / 2_higher	316,833	16,471
edu2	no_hs / has_hs	221,440	16,471
gender	female / male	245,823	16,471

Layer	Layer values	Rows	Unique firms
race	nonwhite / white	230,047	16,448

Built by 06_prep_layer_outcomes.py (edu2) and 06_prep_demog_outcomes.py (gender, race) from worker_panel_lagos.parquet. Each row represents one firm–layer–year cell: all workers at the focal establishment in December of that year who belong to the layer (education bin, gender, or race group). Outcomes are computed as the count and within-cell averages of the relevant worker-level variables.

2.1 Identifiers

Variable	Type	Description
identificad	VARCHAR	14-digit establishment identifier (CNPJ-base). Stored as a zero-padded 14-character string (e.g., '00001961000110'). Join key with firm_layer_connectivity_{layer}.dta.
layer_id	VARCHAR	Layer membership of the workers in this cell. Values depend on the layer: edu2 → {no_hs, has_hs}; gender → {female, male}; race → {nonwhite, white}. Workers with missing layer information are excluded. Together with identificad and year this forms the unique observation key.
year	INTEGER	Calendar year of the December 31 stock. Range: 2009–2016. Only years ≥ 2009 are used in regressions (post-reform data begins 2012).

2.2 Layer Definitions

Layer	Variable	Source field	Mapping
edu	layer_id ∈ {0_no_hs, 1_hs, 2_higher}	grinstrucao	0_no_hs: grinstrucao ∈ {2,3,4,5} (up to complete primary / less than high school). 1_hs: grinstrucao ∈ {6,7} (incomplete or complete secondary). 2_higher: grinstrucao ∈ {8,9} (incomplete or complete higher education). Derived from educ_bin in worker_panel_lagos.parquet using pd.cut(grinstrucao, bins=[0,6,8,11]).
edu2	layer_id ∈ {no_hs, has_hs}	grinstrucao	no_hs: grinstrucao ∈ {2,3,4,5} (up to complete primary). has_hs: grinstrucao ∈ {6,7,8,9} (some secondary or above). Derived by remapping the 3-bin educ_bin variable: 0_no_hs → no_hs; {1_hs, 2_higher} → has_hs.
gender	layer_id ∈ {female, male}	genero	female: genero = 0. male: genero = 1. Workers with missing genero excluded.
race	layer_id ∈ {nonwhite, white}	raca_cor	white: raca_cor = 2 (branca). nonwhite: raca_cor ∈ {8 parda, 4 preta}. Excludes indígena (9), amarela (6), and missing/undeclared. ~5 % of workers are excluded from the race layer for this reason.

2.3 Outcome Variables

All outcomes are computed at the firm–layer–year cell level from worker_panel_lagos.parquet (workers with empem3112 = 1, remdezr > 0, horascontr > 0). Statistics below pool all layers and years for edu2.

Variable	Type	Min	Mean	Max	Nulls	Description
layer_emp	INTEGER	1	84.9 (edu2) 76.5 (gender) 77.7 (race)	11,494 –18,209	0	Count of workers in the layer at this firm in December of the given year (COUNT of PIS values in the cell). Denominator for within-cell averages. Minimum is 1 by construction (cells with zero workers don't appear).

Variable	Type	Min	Mean	Max	Nulls	Description
lr_remdezr_layer	DOUBLE	5.40 (edu2)	7.67 (edu2) 7.80 (gender) 7.81 (race)	11.57	0	Within-cell average of log December earnings deflated to Dec 2015 BRL. Computed as $AVG(lr_remdezr)$ over workers in the cell, where $lr_remdezr = \log(\text{remdezr} / \text{IPCA}_t)$. IPCA deflators: 2009=0.6716, 2010=0.7113, 2011=0.7575, 2012=0.8048, 2013=0.8551, 2014=0.9104, 2015=1.0000, 2016=1.0690.
lr_remdezr_h_layer	DOUBLE	0.61 (edu2)	1.65 (edu2) 1.72 (gender) 1.73 (race)	3.59	0	Within-cell average of log hourly earnings (deflated). Computed as $AVG(\log(\text{remdezr} / \text{horascontr} / \text{IPCA}_t))$ at the worker level before averaging — not equivalent to $lr_remdezr_layer$ minus $\log(\text{hours})$. Workers with $\text{horascontr} = 0$ or $\text{remdezr} = 0$ are excluded.
l_layer_emp	DOUBLE	0.00	2.85 (edu2) 2.74 (gender) 2.71 (race)	9.81	0	Log of layer_emp : $\ln(\text{layer_emp})$. Zero maps to 0 (one-worker cells). Used as the size outcome and as a control in regressions.

Dataset 3 — firm_layer_connectivity_{layer}.dta

Files: firm_layer_connectivity_edu2.dta | firm_layer_connectivity_gender.dta | firm_layer_connectivity_race.dta

Path: Data/layer_connectivity/final_measures/ | **Unit:** establishment × layer (time-invariant) | **Period:** pre-treatment averages over 2007–2011 year pairs

Layer	Rows	Unique firms	Layer values
edu	44,946	17,708	0_no_hs / 1_hs / 2_higher
edu2	31,237	17,708	no_hs / has_hs
gender	33,767	17,708	female / male
race	32,355	17,663	nonwhite / white

Built by 02_aggregate.py / 02d_aggregate_demog.py (year-pair components) and 03_compute_n.py / 03d_compute_n_demog.py (NaN-aware averages). The construction uses worker transition records from transitions_{layer}.parquet. Each observation corresponds to one firm–layer cell and records how connected that cell is to treated firms (those covered by CBAs affected by the Súmula 277 reform) via worker flows in the pre-treatment period.

3.1 Identifiers

Variable	Type	Description
identificad	VARCHAR	14-digit establishment identifier (zero-padded 14-char string, e.g., '00001961000110'). Join key with firm_layer_outcomes_{layer}.dta on (identificad, layer_id).
layer_id	VARCHAR	Layer of the focal firm cell. Same values as in firm_layer_outcomes: edu2 → {no_hs, has_hs}; gender → {female, male}; race → {nonwhite, white}. A firm appears once per layer in which it has at least one worker-year transition record.

3.2 Year-Pair Variables (four per measure)

Each measure below exists as four year-pair columns, labelled *_0708, *_0809, *_0910, *_1011. These are the raw year-pair ratios before averaging. Statistics in the table below are pooled across all year pairs and layers.

Variable stub	Min	Mean	Max	Description
layer_treat_pw_AB	0.000	~0.083–0.120	2.000	ratio_total for year pair AB: (outflows to treated + inflows from treated) at this firm–layer cell, divided by E_avg. This is the layer-specific analogue of totaltreat_pw_n from the firm-level spillover regressions. Numerator counts all treated contacts regardless of the worker's layer at the contact firm.
sametreat_pw_AB	0.000	~0.077–0.111	2.000	ratio_same for year pair AB: M_same / E_avg, where M_same counts treated contacts where the worker's layer at the contact firm equals the focal layer. Captures within-layer transmission of the union treatment: a no_hs worker at a non-treated firm connects to a treated firm through the no_hs layer.
crosstreat_pw_AB	0.000	~0.0005–0.006	2.000	ratio_cross for year pair AB: M_cross / E_avg, where M_cross counts treated contacts where the worker's layer at the contact firm differs from the focal layer. Captures cross-layer transmission: e.g., a has_hs layer at a non-treated firm connecting to a treated firm via a no_hs worker. Very small for gender and large-education splits because most workers maintain their demographic group across firms.
totalflows_layer_pw_AB	0.000	~0.252–0.301	2.000	Total worker flows (to/from any firm, not just treated) for this firm–layer cell in year pair AB, scaled by E_avg. Measures overall labour market attachment of the layer. Used to construct the layer_totalflows_pw_pre4 control bins in regressions.

3.3 NaN-Aware Average Variables (main analysis variables)

Variable	Type	Mean (edu)	Mean (edu2)	Mean (gender)	Mean (race)	Description
layer_treat_pw_n	DOUBLE	0.102	0.083	0.114	0.120	NaN-aware average of layer_treat_pw_AB across the available year pairs. Main treatment variable in 07_layer_spillover.do: scaled to P90 of the untreated balanced-panel subsample at 2009 to form layer_conn_norm.
sametreat_pw_n	DOUBLE	0.091	0.077	0.111	0.111	NaN-aware average of sametreat_pw_AB. Within-layer connectivity component. By construction, sametreat_pw_n + crosstreat_pw_n ≤ layer_treat_pw_n (equality when all treated contacts have an observed contact-firm layer).
crosstreat_pw_n	DOUBLE	0.009	0.004	0.001	0.006	NaN-aware average of crosstreat_pw_AB. Cross-layer connectivity component. Typically much smaller than sametreat_pw_n: most workers who move to a treated firm are in the same demographic group as the focal layer.
totalflows_layer_pw_n	DOUBLE	0.286	0.252	0.285	0.301	NaN-aware average of totalflows_layer_pw_AB. Used to construct the pre-treatment turnover bins (layer_totalflows_pw_pre4) that are interacted with year dummies as additional controls in regressions.
*_count_n	INTEGER	3.7	3.7	3.8	3.7	Number of non-missing year pairs that entered the average for each _n variable. Maximum is 4 (all four year pairs available). Values below 4 arise when the firm-layer is absent from one RAIS cross-section. One _count_n variable is stored per _n variable (e.g., layer_treat_pw_n_count_n).